

# Efficient Malicious Encrypted Traffic Detection via Multi-Scale Convolution-Augmented Transformer: The NetFlowClassifier Approach

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**Abstract**—With the widespread adoption of encryption technologies, an increasing proportion of Internet traffic is encrypted, which fundamentally weakens traditional traffic inspection mechanisms. Existing port-based and payload-based detection methods have become ineffective, while approaches relying on handcrafted features suffer from limited discriminative capability and poor generalization. To address the problem of malicious encrypted traffic identification, this paper proposes a lightweight hybrid model named NetFlowClassifier, together with a targeted data preprocessing pipeline. An encrypted traffic dataset is first constructed from CSE-CIC-IDS2018 for binary classification of benign and malicious flows. The proposed model integrates multi-scale depthwise separable convolution and an improved Transformer encoder to jointly capture local discriminative patterns and long-range dependencies. In addition, learnable feature position encoding and attention-weighted pooling are introduced to enhance feature representation. Experimental results show that NetFlowClassifier achieves an F1-score of 0.9661, outperforming baseline CNN and Transformer models by 1.8%–3.2%. With 11.2 million parameters and a throughput of 128 samples per second, the model effectively balances detection accuracy and computational efficiency. These results demonstrate the effectiveness of the proposed approach for malicious encrypted traffic detection in modern network environments.

**Keywords**—Malicious encrypted traffic identification; Binary classification; Multi-scale depthwise separable convolution; Transformer; Attention-weighted pooling

## I. INTRODUCTION

The rapid adoption of end-to-end encryption and Virtual Private Network technologies has fundamentally reshaped modern network communications. While encryption effectively protects user privacy and data integrity, it also provides a favorable environment for malicious activities, such as ransomware propagation, data exfiltration, and botnet command-and-control communications. An increasing

proportion of advanced cyber attacks are now delivered through encrypted channels, making malicious encrypted traffic detection a critical challenge for network security monitoring.

This paradigm shift has significantly weakened traditional traffic inspection techniques, including port-based classification and Deep Packet Inspection (DPI) [1]. Port-based methods rely on fixed port numbers and are easily bypassed by dynamic port hopping and protocol obfuscation, while DPI becomes ineffective when payload contents are fully encrypted. To address this challenge, early studies introduced handcrafted statistical features combined with traditional machine learning models, such as SVM and Random Forest. However, these approaches heavily depend on domain expertise and exhibit limited generalization capability to emerging encryption protocols and novel attack patterns.

With the emergence of deep learning, convolutional neural networks were introduced to realize end-to-end encrypted traffic classification, with Deep Packet [2] as a representative model. Although CNN-based methods can effectively capture local discriminative patterns, their fixed receptive fields restrict the modeling of long-range dependencies in encrypted traffic sequences. Transformer-based architectures have further improved global feature modeling ability, yet their high computational complexity and insufficient focus on fine-grained malicious characteristics limit practical deployment, especially in resource-constrained network environments. Moreover, many existing studies are evaluated on outdated datasets that fail to reflect modern encrypted attack scenarios, reducing their applicability to real-world networks. In contrast, the CSE-CIC-IDS2018 dataset includes diverse encrypted traffic and contemporary attack types, highlighting the need for more robust and adaptive classification models.

To address these limitations, this paper proposes NetFlowClassifier, a lightweight hybrid deep learning model

for binary classification of encrypted traffic. The main contributions of this paper can be summarized as follows:

- A targeted preprocessing pipeline for CSE-CIC-IDS2018 is designed to filter encrypted flows, normalize 78-dimensional traffic features, and eliminate redundant header noise, thereby enhancing discriminative pattern retention and reducing data heterogeneity.
- A novel hybrid architecture is proposed, integrating multi-scale depthwise separable convolution with an improved Transformer encoder, augmented by learnable feature position encoding and attention-weighted pooling to jointly model local and long-range dependencies.
- A lightweight yet high-performance design is achieved, with 11.2 million parameters and a throughput of 128 samples per second, enabling efficient deployment in resource-constrained edge environments.

## II. RELATED WORK

Early encrypted traffic classification mainly relied on rule-based techniques such as Deep Packet Inspection and port-based mapping, which identify applications through protocol semantics and fixed handshake fields. However, with the evolution of encryption mechanisms and evasion strategies, these approaches have gradually lost effectiveness. The widespread adoption of TLS 1.3, QUIC, and HTTP/3 conceals key handshake features, while port hopping and protocol obfuscation further reduce detection reliability. Consequently, existing rule-based systems show limited adaptability to emerging encrypted protocols and modern malicious traffic.

To overcome the rigidity of rule-based methods, early learning-based approaches introduced handcrafted statistical features combined with traditional machine learning models [3]. These features include entropy, packet size distribution, inter-arrival time, and flow-level attributes, and achieved promising performance on early benchmark datasets. Nevertheless, this paradigm suffers from three inherent limitations: feature design is highly dependent on domain expertise, handcrafted features are insufficient to represent dynamic malicious behaviors, and model generalization across heterogeneous datasets is poor.

Deep learning techniques become a dominant paradigm by enabling automatic feature extraction. Convolutional neural network based models capture local byte-level patterns

effectively, but require reshaping one-dimensional traffic sequences into two-dimensional representations, resulting in increased computational overhead and limited capability in modeling long-range dependencies. To explicitly model temporal correlations among packets, LSTM-based models have been applied to encrypted traffic classification tasks [4]. Although they demonstrate strong performance, these models suffer from poor parallelism, long training time, and gradient degradation on long sequences.

Transformer-based architectures have recently been introduced to model global dependencies in encrypted traffic through self-attention mechanisms [5]. While these models achieve competitive accuracy, their large parameter scale and high computational complexity significantly limit their deployment in resource-constrained network devices. Hybrid architectures that combine convolutional local feature extraction with Transformer-based global modeling have also been explored for encrypted traffic classification [6]. In addition, multi-scale feature extraction strategies and graph neural network based flow modeling have been applied to further enhance encrypted traffic identification performance [7] [8].

Another critical issue concerns dataset timeliness. Most existing studies are evaluated on outdated datasets that lack modern encrypted protocols and sufficient encrypted malicious samples, which fails to reflect current network environments dominated by QUIC, TLS 1.3, and ECH traffic. Recently, large-scale pre-trained and foundation models have emerged as a promising research direction for improving generalization across diverse encrypted traffic scenarios [9].

These observations motivate the design of a more efficient and adaptive classification framework, which will be introduced in the following section.

## III. METHODOLOGY

### A. Model Architecture Overview

To address the insufficient multi-scale feature representation and inefficient long-range dependency modeling in encrypted traffic classification, we propose NetFlowClassifier, a lightweight hybrid framework that integrates structured feature embedding, multi-scale local feature extraction, and global dependency modeling in an end-to-end manner.

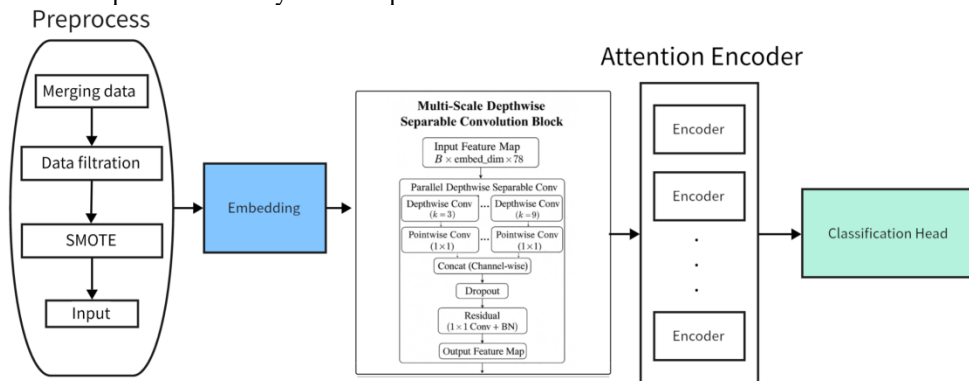


Fig. 1. Overall architecture of the proposed NetFlowClassifier.

As illustrated in Fig. 1, the model follows an embedding–enhancement–sequence modeling–adaptive pooling–classification pipeline. Given 78-dimensional structured NetFlow features, NetFlowClassifier sequentially applies a multi-scale convolution module, a stacked Transformer encoder, and an attention-weighted pooling layer to obtain discriminative global representations for malicious/benign classification. The hybrid design enables the model to jointly capture fine-grained local patterns and long-range feature dependencies while maintaining lightweight characteristics suitable for edge deployment.

### B. Data Preprocessing Module

To construct a high-quality encrypted traffic dataset, a targeted preprocessing pipeline is designed based on CSE-CIC-IDS2018. Encrypted flows are first filtered using protocol and port-based rules. Redundant, non-numeric, and privacy-sensitive fields are removed, and all features are normalized into the range of [0, 1].

To mitigate severe class imbalance, a batch-based SMOTE strategy is adopted to synthesize minority-class samples, yielding a balanced dataset suitable for supervised training. This preprocessing process ensures that discriminative encrypted malicious patterns are retained while reducing data heterogeneity.

### C. Multi-Scale Depthwise Separable Convolution Module

To capture local discriminative patterns of different granularities, a MultiScaleConvBlock is designed using depthwise separable convolution for computational efficiency. Four parallel depthwise convolution kernels with sizes 3, 5, 7, and 9 are applied to extract multi-scale local features, followed by pointwise convolution for channel fusion. Residual connections and normalization are employed to stabilize training.

This module enables the model to capture both fine-grained and coarse-grained encrypted traffic patterns while significantly reducing computational complexity compared to standard convolution.

### D. Transformer Encoder Module

To strengthen long-range dependency modeling while controlling computational cost, the NetFlowClassifier adopts a stacked Transformer encoder architecture with four layers and multiple optimizations, including pre-normalization, reinforced residual connections, and 8-head attention mechanisms. Fig. 2 illustrates the structure of the Transformer encoder in the NetFlowClassifier Model.

The Multi Head Attention module adopts a Transformer-XL-style pre-normalization strategy, applying layer normalization before attention computation to stabilize deep network training, with an input dimension of 128 and 8 attention heads to balance feature resolution and efficiency. The multi-head mechanism partitions the feature space into subspaces, enabling simultaneous attention to different dependency types, while residual connections preserve low-level features by directly adding the attention output to the original input. The Feed\_Foward module incorporates a "pre-normalization + bottleneck layer" structure, with a hidden dimension of 512 to

expand the feature space and capture complex non-linear relationships, followed by dual dropout layers to prevent overfitting. The sequential encoding process of the four stacked layers progressively refines feature representations, building on multi-scale local features and attention-based global dependencies to model complex relationships such as correlations between TLS handshake features and subsequent data transmission patterns. The attention mechanism is defined as:

$$Attention(Q^h, K^h, V^h) = Softmax\left(\frac{Q^h K^{hT}}{\sqrt{16}}\right)V^h \quad (1)$$

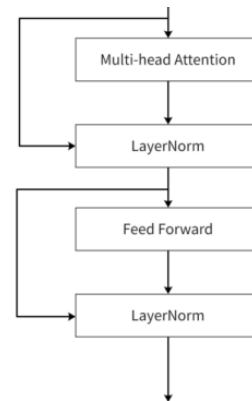


Fig. 2. Internal Structure Schematic of the Transformer Encoder in the NetFlowClassifier Model.

### E. Feature Enhancement and Classification Module

The classification module integrates learnable feature position encoding, attention-weighted pooling, and a hierarchical classification head, combined with advanced training strategies to convert high-dimensional sequence features into discriminative class probabilities. The FeaturePositionEncoding module, tailored for 78-dimensional structured features, introduces a trainable position embedding matrix and a feature importance scaling vector to dynamically adjust the weight of each feature dimension, enabling the model to learn the semantic importance of different feature positions and emphasize discriminative indicators. The AttentionPooling module replaces traditional mean pooling with a two-layer neural network to compute attention weights for each of the 78 feature positions, prioritizing high-contribution features through softmax normalization and weighted summation to generate a compact 128-dimensional global feature vector. The hierarchical classification head adopts three linear layers with multi-layer normalization and GELU activation, progressively compressing feature dimensions while stabilizing feature distribution and capturing complex decision boundaries. To improve model robustness and training stability, a custom LabelSmoothingLoss function replaces standard cross-entropy loss, the AdamW optimizer with weight decay mitigates overfitting, and a CosineAnnealingLR scheduler decays the learning rate from 0.0005 to 1e-6 over 50 epochs to balance exploration and convergence.

### F. Algorithm Description of NetFlowClassifier

To improve the clarity and reproducibility of the proposed framework, this subsection summarizes the overall training and

inference procedure of NetFlowClassifier in the form of pseudo-code. The algorithm describes how structured NetFlow features are embedded, enhanced by multi-scale convolution, encoded by the Transformer layers, aggregated through attention-weighted pooling, and finally mapped to classification results. It explicitly illustrates the execution order of key modules and the data flow among them, facilitating reproducibility and practical implementation. The Fig.3 shows the algorithm description of NetFlowClassifier.

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**Algorithm 1: Training and Inference Procedure of NetFlowClassifier**

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**Input:** Encrypted traffic feature matrix  $X \in R^{N \times 78}$   
**Output:** Class label  $y \in \{0,1\}$

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1: Normalize  $X$  into  $[0,1]$ 
2: for each mini-batch  $B$  in  $X$  do
3:   // Feature Embedding
4:    $H0 \leftarrow \text{LinearEmbedding}(B)$ 

5:   // Multi-scale convolution
6:   for  $k \in \{3, 5, 7, 9\}$  do
7:      $Hk \leftarrow \text{DepthwiseConv}(H0, \text{kernel}=k)$ 
8:   end for
9:    $Hm \leftarrow \text{Concat}(H3, H5, H7, H9)$ 
10:   $Hm \leftarrow \text{PointwiseConv}(Hm) + H0$  // residual

11:  // Transformer Encoder
12:  for  $l = 1$  to  $L$  do
13:     $Hl \leftarrow \text{LayerNorm}(Hm)$ 
14:     $Hl \leftarrow \text{MultiHeadAttention}(Hl)$ 
15:     $Hl \leftarrow Hl + Hm$ 
16:     $Hl \leftarrow \text{FFN}(Hl)$ 
17:  end for

18:  // Feature Position Encoding
19:   $Hp \leftarrow \text{FeaturePositionEncoding}(Hl)$ 

20:  // Attention Pooling
21:   $\alpha \leftarrow \text{Softmax}(W2 \cdot \tanh(W1 \cdot Hp))$ 
22:   $g \leftarrow \sum \alpha \odot Hp$ 

23:  // Classification
24:   $\hat{y} \leftarrow \text{MLP}(g)$ 
25: end for
26: return  $\hat{y}$ 

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Fig. 3. The algorithm description of NetFlowClassifier.

## IV. EXPERIMENTS

### A. Dataset Preparation

This study evaluates NetFlowClassifier on an encrypted traffic dataset derived from CSE-CIC-IDS2018. Encrypted flows are first filtered using protocol and port-based rules, and redundant or non-numeric features are removed. To alleviate severe class imbalance, a batch-based SMOTE strategy is applied, resulting in a balanced dataset of 2,659,325 samples, including 2,005,733 benign and 653,592 malicious flows. The dataset is then divided into training, validation, and testing subsets with an 8:1:1 stratified split. Detailed information on the data partition is presented in Table 1.

TABLE I. DATASET PARTITION OF THE CSE-CIC-IDS2018 ENCRYPTED TRAFFIC DATASET

| Dataset         | Training Set | Validation Set | Testing Set | Total     |
|-----------------|--------------|----------------|-------------|-----------|
| CSE-CIC-IDS2018 | 2,127,460    | 265,933        | 265,932     | 2,659,325 |

### B. Experimental Environment and Hyperparameters Settings

In terms of training hyperparameters, the model was trained for a total of 50 epochs with a mini-batch size of 128 to balance gradient stability and training speed. The initial learning rate was

set to 0.0005, and the AdamW optimizer was adopted with weight decay of 0.01 and betas (0.9, 0.999) to suppress overfitting; a CosineAnnealingLR learning rate scheduler was applied to dynamically adjust the learning rate, with the minimum learning rate decayed to 1e-6 over the training period to facilitate late-stage convergence.

A dropout rate of 0.15 was uniformly applied to key modules (embedding layer, multi-head attention, FFN, and classification head) to enhance model generalization, and label smoothing with a smoothing factor of 0.1 was introduced to the cross-entropy loss function to alleviate over-confidence in hard labels.

To prevent overfitting, an early stopping strategy was implemented with a patience of 7 epochs: if the validation accuracy did not improve by more than 1e-6 for 7 consecutive epochs, training was terminated early, and the model checkpoint with the highest validation accuracy was saved. Additionally, all feature inputs were normalized to the range [0,1] before feeding into the model, and the position encoding module adopted learnable parameterization to adapt to the structural characteristics of NetFlow features.

### C. Baseline Models and Metrics

To evaluate the effectiveness of NetFlowClassifier, three representative baselines are selected:

- SVM: Traditional machine learning model using original NetFlow features.
- Basic1DCNN: Single-scale convolution-based model focusing on local feature extraction.
- ResNet\_Model: Residual 1D-CNN model enhancing feature propagation.

Performance is evaluated using Accuracy, Precision, Recall, and F1-score, along with parameter size and inference throughput to assess deployment efficiency.

### D. Performance Comparison

Table 2 summarizes the performance of all models. NetFlowClassifier achieves the best overall performance with an Accuracy of 96.61% and F1-score of 0.9652, outperforming Basic1DCNN and ResNet\_Model by 1.49% and 0.49% in accuracy, respectively. It also significantly improves malicious traffic detection capability compared to all baselines.

TABLE II. PERFORMANCE COMPARISON OF DIFFERENT MODELS

| Model                    | Accuracy (%) | Weighted Precision | Weighted Recall | Weighted F1-score |
|--------------------------|--------------|--------------------|-----------------|-------------------|
| SVM                      | 85.55        | 0.8716             | 0.8555          | 0.8339            |
| Basic1DCNN               | 95.12        | 0.9531             | 0.9512          | 0.9515            |
| ResNet_Model             | 96.12        | 0.9627             | 0.9612          | 0.9601            |
| <b>NetFlowClassifier</b> | <b>96.61</b> | <b>0.9674</b>      | <b>0.9661</b>   | <b>0.9652</b>     |

These performance advantages stem from the model's core innovations: the multi-scale depthwise separable convolution module captures local discriminative features of varying granularities; the improved Transformer encoder strengthens the extraction of long-range feature dependencies; and the learnable

feature position encoding and attention-weighted pooling enable dynamic focus on key malicious indicators.

Despite its hybrid architecture, the NetFlowClassifier maintains a lightweight design with only 11.2 million parameters—achieved through depthwise separable convolution and optimized Transformer layers. It delivers an inference throughput of 128 samples per second.

#### E. Ablation Experiment

An ablation experiment is conducted by removing the multi-scale convolution module while retaining the Transformer encoder and feature enhancement components. The results of the ablation experiment are presented in Table 3.

TABLE III. THE RESULTS OF THE ABLATION EXPERIMENT

| Model                         | Benign<br>(Precision/<br>Recall/F1-<br>score) | Malicious<br>(Precision/<br>Recall/F1-<br>score) | Overall<br>Accuracy<br>(%) | Weighted<br>F1-score |
|-------------------------------|-----------------------------------------------|--------------------------------------------------|----------------------------|----------------------|
| Transformer-<br>only          | 0.9526/0.99<br>38/0.9728                      | 0.9817/0.84<br>29/0.9076                         | 95.74                      | 0.9576               |
| <b>NetFlowClas<br/>sifier</b> | <b>0.9566/0.99<br/>96/0.9780</b>              | <b>0.9997/0.86<br/>31/0.9259</b>                 | <b>96.61</b>               | <b>0.9652</b>        |

The ablated model shows a noticeable performance degradation, with overall accuracy decreasing to 95.74% and weighted F1-score dropping to 0.9576, confirming that multi-scale convolution plays a critical role in capturing fine-grained malicious traffic patterns that cannot be sufficiently modeled by self-attention alone.

#### V. CONCLUSION

This paper proposes NetFlowClassifier, a lightweight hybrid deep learning framework for malicious encrypted traffic identification. By integrating multi-scale depthwise separable convolution with an improved Transformer encoder, together with learnable feature position encoding and attention-weighted pooling, the proposed model effectively addresses the limitations of existing approaches in multi-scale feature extraction and long-range dependency modeling.

Experiments conducted on the processed CSE-CIC-IDS2018 encrypted traffic dataset demonstrate that NetFlowClassifier achieves an F1-score of 0.966, outperforming representative CNN- and Transformer-based baselines while maintaining a lightweight architecture with only 11.2 million parameters and an inference throughput of 128 samples per second. These results confirm that the proposed model provides a favorable

balance between detection accuracy and deployment efficiency, making it suitable for resource-constrained edge environments.

Future work will focus on extending the dataset to include emerging encrypted protocols and real-world traffic traces, improving model interpretability through attention visualization and feature attribution, and further exploring multi-class attack classification and federated learning frameworks to enhance privacy preservation and cross-domain generalization.

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